MIL-R-15058G(SHIPS)
18 February 1965
SUPERSEDING
MIL-R-15058F(SHIPS)
25 January 1960
(See paragraph 6.7)

MILITARY SPECIFICATION

RUBBER, SHAFT COVERING MATERIALS

(FOR MARINE PROPELLER SHAFTS)

1. SCOPE

1.1 Scope. - This specification covers materials and techniques used to protect propeller shafts from corrosion.

1.2 Classification. - The shaft covering shall be of the following types, as specificed (see 6.1):

Type II - Sheet material and associated cements and tie gums which can be vulcanized to solid or hollow shafts for shop applications and new shafts.

Type III - Prevulcanized sheet material, primer, and associated cements which can be adhered to either hollow or solid shafts for dry dock applications.

Type IV - Brushing compound and associated primers, cements and troweling material which can be vulcanized to either hollow or solid shaft couplings for dry dock applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.

MILITARY
MIL-B-16541 - Bronze, Valve: Castings.
MIL-D-16791 - Detergents, Liquid, Nonionic.
MIL-S-23284 - Steel, Forgings, Carbon and Alloy, For Shafts, Sleeves, Couplings and Stocks (Rudders and Diving Planes).

STANDARD

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5330

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2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material. - The materials in types II, III and IV shaft coverings shall be synthetic rubber, chloroprene, butadiene-styrene, or butediene-acrylonitrile, that conform to the applicable requirements specified in table I.

Table I - Physical requirements of shaft coverings.

		All types	Test procedure
Tests on covered shafts:			
Porosity test		no porosity	4.4.1
Hardness, durometer points		50 to 80	4.4.2
Tests on covered test jigs:			
Corrosion test		no corrosion	4.4.3
Adhesion to steel shaft, metal sleeves, and between plies of rubber, rate of stripping under 20 pound load, inches per minute, maximum		1	4.4.4
	Type II	Type III or IV	
Tensile strength, psi, minimum	1800	1200	4.4.5
Ultimate elongation, percent, minimum	300	250	4.4.5
		All types	
Tensile strength after oven aging, percent of initial, minimum		80	4.4.6
Volume change after water immersion, percent		0 to 15	4.4.7
		Type IV only	
Impact resistance		No cracking, shattering, or failure	4.4.8
Thermal shock resistance		No cracks or defects	4.4.9

3.2 Preparation of shaft and test jigs.-

3.2.1 Bronze or monel bearing sleeves. - Bronze or monel bearing sleeves, on either end of the surfaces to be covered, shall have a tapered undercut surface as shown on figures 1 and 2, as applicable for the type of shaft covering. These sleeves may be coated with another metal in order to obtain the adhesion as specified in 3.1.

3.2.2 Steel shafts. - Steel shafts and sleeves of bearing, in the areas to be covered, shall be prepared by cleaning the surface either by sandblasting or gritblasting. Immediately after cleaning, the surface shall be covered with a suitable primer and cement. While the cement is drying it shall be protected from dust or dirt until ready to proceed with application of the covering.

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3.2.3 Test jigs to represent the shaft and the bearing sleeves. - Two test jigs of the metal in the shaft and two test jigs of that in the bearing sleeves will be required for each shaft. Unless otherwise specified (see 6.1), the test jigs shall be supplied by the contractor. The test jigs shall be at least 4 inches in diameter and at least 6 inches long. The jigs may be suitable pipe sections to represent either solid or hollow shafts. The ends of the pipe sections shall be closed except for suitable pipe connections. The steel test jigs representing the shaft shall conform to MIL-S-23284 and the bronze or monel test jigs, as applicable, representing the bearing sleeves shall conform to MIL-B-16541 or QQ-N-281, respectively. The test jigs shall be covered at the same time as the shaft and bearing sleeves, using the same materials and techniques.

3.3 Application of shaft coverings. -

3.3.1 Type II. - Unvulcanized calendered sheets shall be applied to give a finished thickness of 3/16 inch, plus 1/16 or minus 1/32 inch. Over-lapping edges of each sheet shall be beveled to fair joints. No cements shall be used between the sheets. Precautions shall be taken to insure that air is not trapped between the surface of the shaft and the synthetic rubber sheets, or between the layers of synthetic rubber. The rubber covering may be thicker than 3/16 inch where shaft surfaces are to be protected between the ends of shaft sleeve materials in stern tubes. In this case the rubber covering must be the same thickness as the shaft sleeves.

3.3.1.1 Vulcanizing process. - The vulcanizing process shall consist of curing the covered shaft and test jigs simultaneously in an atmosphere of live steam at atmospheric pressure or by any other method which will give sufficient heat to accomplish the same results. For hollow shafts, the synthetic rubber covering may be vulcanized by blowing steam through the bore of the shaft and the test jigs. In this case, test jigs representing shaft coverings shall be connected in series with the inlet or outlet steam used in vulcanizing the shaft covering. After vulcanization, the test jigs shall be subjected to the tests specified in 4.4.

3.3.2 Type III. - A minimum of two plies of vulcanized, roughened sheets shall be applied in such a manner as to give a finished thickness of 1/8 inch plus or minus 1/32 inch. Overlapping edges of each sheet shall be beveled to fair joints. Joints of the covering shall be staggered, preferably 180 degrees from each other. Suitable cements shall be used between sheets and may be used over the outer sheet. Precautions shall be taken to insure that neither solvent nor air is trapped between the surfaces of the shaft and the sheets, or between the plies. Test jigs, prepared at the same time, shall be subjected to the tests specified in 4.4.

3.3.2.1 Type III material for stock. - When the covering material is furnished for stock, the sheet shall be roughened by griding on both sides. Complete instruction for application shall be furnished. These instructions shall include the required metal surface preparation, application of component materials, mixing of two part materials, drying time of primers and cements, and recommended curing conditions of time and temperature.

3.3.3 Type IV. - Synthetic rubber brushing compound and troweling compound applied as specified by the manufacturer shall have a final thickness of 0.040 to 0.060 inch. The covering shall be anchored at the edges by filling the undercut grooves of the bearing sleeves as shown in figure 2.

3.3.3.1 Vulcanization of Type IV covering. - The vulcanizing process shall consist of curing the shaft covering and the test jigs simultaneously in an atmosphere of hot air or by any other method which will give sufficient cure. After vulcanization, the test jigs shall be subjected to the tests specified in 4.4.

3.3.3.2 Type IV covering for stock. - When the covering material is furnished for stock, curing accelerators shall be furnished in quantities suitable to cure the materials involved and complete instructions shall be included. These instructions shall include the required surface preparation, application of component materials, mixing of two part materials, drying time of materials, and recommended curing conditions of time and temperature.

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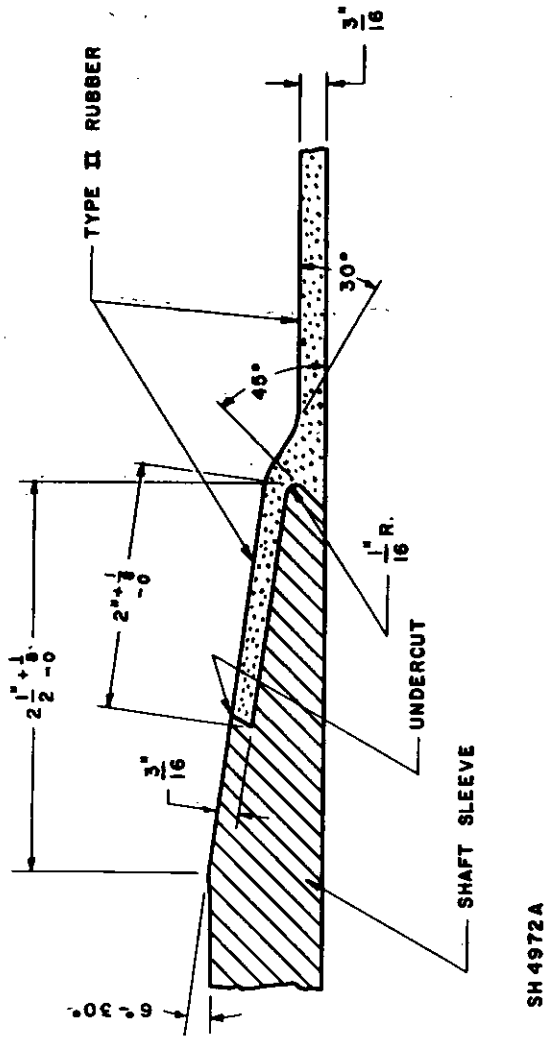


Figure 1 - Taper and undercut on bronze or monel sleeves for type II shaft covering.

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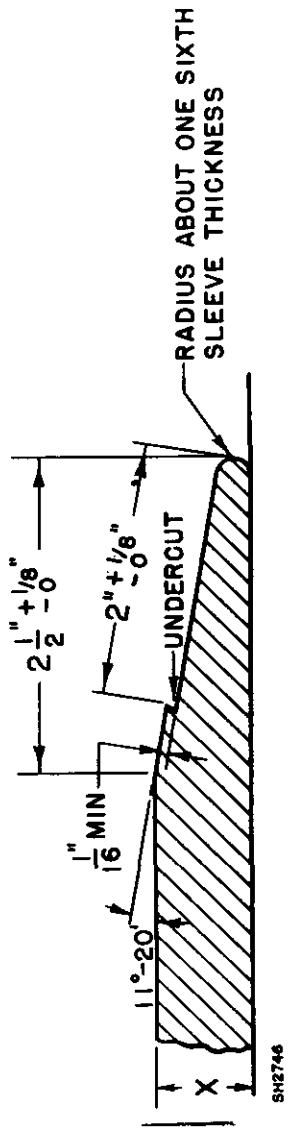


Figure 2 - Taper and undercut on bronze for types III and IV shaft covering.

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3.4 Finished shaft coverings. - The finished shaft coverings shall be resistant to salt water, shall have adequate adhesion to steel shafts and bronze or monel bearing sleeves, and shall afford adequate protection against corrosion, (see 3.1).

3.5 Workmanship. - All shaft coverings shall be free from blisters, pits, voids, trapped air, tears, open seams, or other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance examination and tests on finished shaft coverings. -

4.2.1 Visual examination of finished shaft covering. - Each finished shaft covering shall be examined for workmanship (see 3.5). If any defects are found, they must be suitably repaired before acceptance of the shaft covering.

4.2.2 Tests on finished shaft covering. - Each finished shaft covering shall be tested for porosity and hardness as specified in 4.4.1 and 4.4.2, respectively.

4.2.2.1 Action in case of nonconformance. - If the covering on the shaft does not conform to the hardness requirement of 3.1, this shall be cause for rejection. If the covering has porosity, it shall be suitably repaired and rechecked for porosity.

4.2.3 Tests on jigs representing covered shafts. - Quality conformance tests on the test jigs representing covered shafts shall consist of the tests specified in 4.4.2 through 4.4.9, as applicable.

4.2.3.1 Rejection. - If any sample of types II, III, or IV shaft covering as applied to a test jig fails to meet the applicable requirements for the tests specified in 4.2.3, this shall be cause for rejection of the finished shaft covering represented by the test jig.

4.3 Material furnished for stock (types III and IV only). -

4.3.1 Sampling. -

4.3.1.1 Lot. - All the material of the same type, produced in one plant under essentially the same conditions, and offered for delivery at one time shall be considered a lot for purposes of sampling and testing.

4.3.1.2 Sampling procedure. - From each lot sufficient material shall be taken in the proportions specified by the manufacturer, to provide for the covering of a 2-square-foot area approximately 1/8 inch thick for type III and 0.040 to 0.060 inch thick for type IV.

4.3.2 Tests on material for stock. -

4.3.2.1 Preparation of test specimens. - Two specimens 12 by 12 inches in size shall be prepared by applying the shaft covering in accordance with manufacturer's instructions on 1/8-inch mild steel plates to the thickness specified. The type IV specimens shall be cured as recommended by the manufacturer before tests are conducted.

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4.3.2.2 Tests on specimens prepared from stock material. - The specimens prepared as specified in 4.3.2.1 shall be subjected to the tests specified in 4.4.2 through 4.4.9, as applicable.

4.3.3 Rejection (types III and IV only). - If a sample fails to meet the requirements for any of the tests specified in 4.3.2.2, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures. - No tests shall be conducted until at least 48 hours after vulcanizing to allow the shaft to cool completely. This applies to types II and IV. Type III covering shall be tested not less that 96 hours after application. The shaft coverings shall be tested by using test jigs or specimens from test jigs representing the finished shaft covering for the tests specified in 4.4.2 to 4.4.7 inclusive.

4.4.1 Porosity tests on finished shaft coverings. -

4.4.1.1 High-frequency spark test on types II and III only. - The finished shaft coverings of types II and III shall be tested electrically for porosity with a high-frequency, high-voltage spark tester delivering a minimum of 10,000 volts. The presence of a pinhole in the covering will be indicated by a spark jumping from the electrode of the tester to the hole in the covering. The corona from the electrode of the tester must not be confused with a spark, which may be distinctly seen and heard. Pinholes located in this fashion shall be repaired to the satisfaction of the inspector before the shaft covering will be considered acceptable.

4.4.1.2 Electrical conductivity test on type IV shaft covering only. - Type IV shaft covering shall be tested for porosity by means of electrical conductivity. The coated surfaces shall be wetted with an aqueous solution containing by volume 1 percent sodium chloride and 0.1 percent of detergent conforming to MIL-D-16791. The swab (cotton pad or sponge) shall be saturated with the solution to assure good electrical conductivity during the test. By means of electrical contact with an uncoated surface of the shaft, the metal base shall be the cathode of a circuit which includes a 90-volt "B" battery and suitable instrumentation. The circuit shall be completed through a flexible lead to the saturated swab. The coated surface shall be searched completely for porosity. A metered change in the electrical circuit (increase in current rate or decrease in voltage) indicates porosity in the covering. The area under the swab shall be dried and marked. Porosity located in this fashion shall be repaired to the satisfaction of the inspector before the shaft covering will be considered acceptable. The K-D Bird Dog[1/] has been found to be a suitable alternate for electrically testing the continuity of shaft coverings.

4.4.2 Hardness. - Hardness shall be determined on the shaft covering of test jigs representing covered shafts and on the coverings of the finished shafts. A 3-second reading shall be taken using a Shore A durometer in accordance with method 3021 of FED-STD-601.

4.4.3 Corrosion test. - A sample test jig (pipe section) representing each finished shaft covering shall be immersed in 3 percent sodium chloride solution at 27° plus or minus 5°C. (80° ± 9°F) for 144 ± 1 hours. The test jig shall then be removed from the salt water and the coating removed from the jig to determine if any corrosion has taken place under the covering.

4.4.4 Adhesion. - The adhesion between the rubber and the test jigs representing each metal of the finished shaft covering shall be determined at room temperature. The adhesion between the plies of type III rubber sheet material shall be determined also. One inch wide strips shall be marked off on the test jigs and a groove cut around the strips so that the strips are free from the remaining shaft covering. The test jig shall be supported at the ends in a horizontal position. One end of each strip shall be separated from the jig for a distance of about three inches. The test jig shall be rotated until the strip is at the bottom in a horizontal position. A dead-weight load of 20 pounds (clamp plus weight) shall be suspended

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[1/] K-D Co., 2091 Cornell St., Palo Alto, Calif. or Gardner Laboratory, Inc., 5521 Landy Lane, Bethesda 14, Maryland.

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from the free end of the rubber strip. The rate of stripping shall be noted in inches per minute. Three strips for each test condition shall be evaluated and the results averaged for each. The rubber strips removed from the jigs shall be examined for imperfections, blisters, voids, delamination, or porosity. If excessive defects are found in the strips, it shall be cause for rejection of the covered shaft represented by the test jigs.

4.4.5 Tensile strength and ultimate elongation. - Tensile strength and ultimate elongation of the rubber covering shall be determined by methods 4111 and 4121, respectively, of FED-STD-601. Die III specimens shall be used.

4.4.6 Oven aging. - Tensile test specimens shall be oven aged in accordance with method 7221 of FED-STD-601 except that the aging shall be 96 plus or minus 1/2 hours at 70° plus or minus 1°C (158° ± 2°F).

4.4.7 Volume change. - Volume change after water immersion of the rubber covering shall be determined in accordance with method 6211 of FED-STD-601 except that the immersion shall be 96 plus or minus 1 hours in boiling water.

4.4.8 Impact test (type IV only). -

4.4.8.1 Specimen. - The specimen to be tested shall be the same specimen as used for the porosity test specified in 4.4.1.2. The specimen shall have been prepared for 24 plus or minus 5 hours, prior to the time the impact test is made.

4.4.8.1.1 Procedure. - The test shall be made by dropping a 1 plus or minus 1/16 inch diameter steel ball, weighing 67 plus or minus 5 grams, from a height of 20 inches. The test shall be performed 10 times at various places on the surface of the specimen at 60° plus or minus 5°F. (15.56° ± 2.8°C.). This test shall be repeated at 0° plus or minus 5°F. (minus 17.78° ± 2.8°C.) dropping the ball from 5 inches. In each case the shaft and covering shall be conditioned for 3 hours at the test temperature prior to test. After all impacts have been made the covering shall be examined visually and by the porosity test (see 4.4.1.2) for any cracking, shattering or failure.

4.4.9 Thermal shock test (type IV only). -

4.4.9.1 Specimen. - The specimen to be tested shall be a test section prepared as specified in 4.4.4.

4.4.9.1.1 Procedure. - Within 24 hours after preparation, the test section shall be placed in a constant temperature box, the temperature of which shall be held at 0° plus or minus 5°F. (minus 17.78° ± 2.8°C.) and left there for not less than 7 nor more than 17 hours. The test section shall then be removed and immediately placed in a constant temperature oven held at a temperature of 120° plus or minus 5°F. (48.89° ± 2.8°C.) for 12 plus or minus 5 hours. The test section shall then be removed to the cold box and the cycle repeated. The alternate cooling and heating cycle shall be repeated six times. At the end of the sixth cycle, the covering on the test section shall be examined visually and by the electrolysis test specified in 4.4.1.2 for any cracks or defects.

4.5 Examination of preparation for delivery. - Sample packs shall be examined to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use. -

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5.1.1 Packaging. - Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transporation.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination, and precautionary warning labels as required for hazardous chemicals.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made at a Government activity (see 6.1).

5.2.1.1 Packaging. -

5.2.1.1.1 Types III and IV. -

5.2.1.1.1.1 Level A. -

5.2.1.1.1.1.1 Sheet material shall be furnished in a minimum of 36-inch wide rolls and a minimum of 25 yards long. Rolls shall be unit packaged in fiberboard boxes conforming to PPP-B-636. Box closure shall be as specified in the appendix to the box specification.

5.2.1.1.1.1.2 Priming cement and brushing or spraying compound shall be furnished in 5 gallon pails conforming to type II, class 1 or 2 of PPP-P-704, or 1 gallon cans conforming to type V, class 2 of PPP-C-96. Exterior plan B coating and side seam stripping of cans shall be required. Cans shall be provided with wire handles.

5.2.1.1.1.1.3 Cement shall be furnished in one gallon cans in accordance with 5.2.1.1.1.1.2.

5.2.1.1.1.1.4 Troweling material shall be furnished in 1 pound cans in accordance with 5.2.1.1.1.1.2.

5.2.1.1.1.1.5 Curing accelerator shall be packaged in accordance with MIL-P-3863 for overseas shipment.

5.2.1.1.2 Types II, III, and IV. -

5.2.1.1.2.1 Level C. - Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity for early installation. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2.1.2 Packing. -

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5.2.1.2.1 Level A.- Sheet material, packaged as specified (see 6.1), shall be packed in overseas type wood-cleated fiberboard, nailed wood, fiberboard, wirebound wood, wood-cleated veneer paper overlaid, or wood-cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-636 (weather-resistant class), PPP-B-585, PPP-B-576 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes may be banded with tape conforming to type IV of PPP-T-97 and the appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.1.2.1.1 Priming cement and brushing or spraying compound furnished in 5 gallon pails shall not require additional packing.

5.2.1.2.1.2 Cement and troweling material packaged as specified in 5.2.1.1.1.1.3 and 5.2.1.1.1.1.4, respectively, shall be packed for overseas shipment in accordance with appendix of PPP-C-96.

5.2.1.2.1.3 Curing accelerator packaged as specified in 5.2.1.1.1.1.5 shall be packed in accordance with overseas shipment of MIL-P-3863.

5.2.1.2.2 Level B.- Sheet material, packaged as specified (see 6.1), shall be packed in domestic type wood-cleated fiberboard, nailed wood, wirebound wood, cleated-plywood or wood-cleated veneer paper overlaid boxes or weather-resistant class fiberboard boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, PPP-B-576 or PPP-B-636, respectively, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto, except strapping of fiberboard boxes shall not be required. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable fiberboard box specification.

5.2.1.2.2.1 Priming cement and brushing or spraying compound furnished in 5 gallon pails shall not require additional packing.

5.2.1.2.2.2 Cement and troweling material packaged as specified in 5.2.1.1.1.1.3 and 5.2.1.1.1.1.4, respectively, shall be packed for domestic shipment in accordance with appendix of PPP-C-96.

5.2.1.2.2.3 Curing accelerator packaged as specified in 5.2.1.1.1.1.5 shall be packed for domestic shipment in accordance with MIL-P-3863.

5.2.1.2.3 Level C.- Sheet material, packaged as specified (see 6.1), shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner which will insure carrier acceptance and safe delivery to destination. Shipping containers shall comply with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transporation.

5.2.1.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked with the date (month and year) of manufacture, nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination and in accordance with MIL-STD-129.

5.2.1.3.1 Hazardous chemicals.- All packages containing hazardous chemicals shall have affixed thereto such warning labels and markings as may be required by the Interstate Commerce Commission Regulations, CFR Title 49, parts 71-78 and the Manufacturing Chemists Association Manual L-1.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) State if test jigs are not required by contractor (see 3.2.3).
(d) Packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 Type II covering may be applied to the shafts at the plant of the contractor or that of the purchaser. Where the covering is applied at the latter, the purchaser shall prepare the shafts for the synthetic rubber covering and shall furnish steam at the pressure required by the contractor, ship facilities including lathes, cranes, shaft supports, and pipe connections with gages and thermometers and shall assist the contractor in every practicable way. An agreement as to procedure and requirements should be reached with the contractor as early as possible.

6.3 The following is the equipment that will probably be required:

(a) A lathe for holding the shafts while the synthetic rubber is being applied.

(b) A table for cutting and trimming the synthetic rubber sheets. This table should be of 2 inch planking and should be 2 feet longer than the longest section of synthetic rubber which is to be put on the shaft. The table should be 45 inches wide. The top surface and the 2 sides should be planed smooth, free from slivers. The table should be approximately 33 inches high.

(c) A pair of wooden horses, the same height as the table, for supporting the rolls of synthetic rubber compound while unrolling the synthetic rubber stock onto the table.

(d) Heavy horses or supports for each shaft while it is being vulcanized. Usually 2 shafts are vulcanized in parallel. These supports should be high enough so that one can easily see all surfaces of the shafts while they are resting thereon.

(e) A double hook traveling crane to move the shafts from the lathe to the vulcanizing location. Until these covered shafts are thoroughly vulcanized and cooled, no lifting tackle of any kind can be placed on the synthetic rubber surfaces.

(f) Steam connections for supplying steam to the shafts and connections for removing the condensate. A pressure gage and thermometer should be installed at the inlet and outlet connections.

6.4 The following labor will probably be required: One or two men to be constantly with the contractor's employees to do pipe fitting, operate lathe, and do rigging.

6.5 Safety measures.- The materials used for synthetic rubber covering are inflammable. No open flames, sparks, or very hot cutting tools should be allowed to operate within 50 feet of these materials. Fire extinguishers of the CO_2 type should be available at the work.

6.6 Type I material is no longer in use and has been deleted from the specification.

6.7 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

D14000

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